Unit 8 Pre-Test

Coordinate Geometry

Standards assessed: NC.M1.G-GPE.4 · G-GPE.5 · G-GPE.6

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.G-GPE.4 · NC.M1.G-GPE.5 · NC.M1.G-GPE.6 Find midpoints, slopes, and lengths on the coordinate plane.

• *I can find a midpoint, a slope, a distance, and the slopes of parallel and perpendicular lines.*

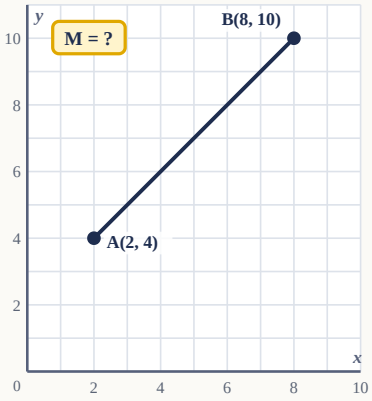
Answer each question. Use the picture to help you.

How to type your answers

point `(5, 7)` fraction `-3/2` root `sqrt(20)` equation `y=2x-2`

1. *Midpoint*

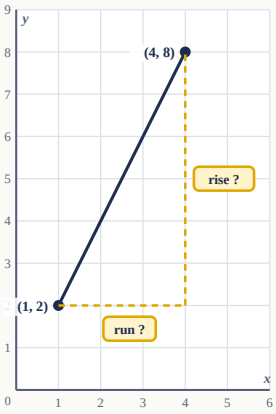
Find the midpoint M of A(2, 4) and B(8, 10).



M = _____

2. *Slope*

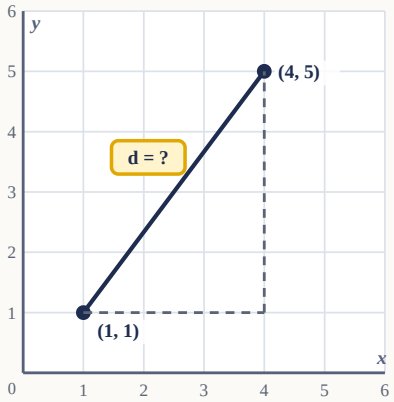
Find the slope of the line through (1, 2) and (4, 8).



m = _____

3. *Distance*

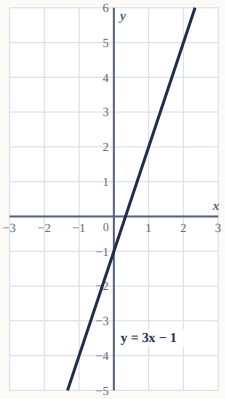
Find the distance between (1, 1) and (4, 5).



d = _____

4. *Parallel slope*

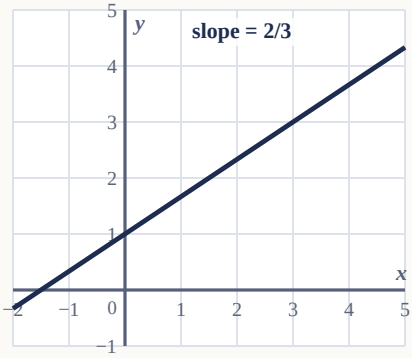
A line is parallel to y = 3x − 1. What is its slope?



m = _____

5. *Perpendicular slope*

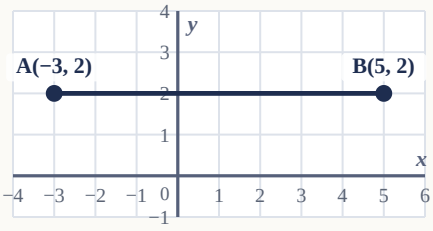
A line has slope $\frac{2}{3}$. What is the slope of a line perpendicular to it?



m = _____

6. *Length*

Find the length of segment AB with A(−3, 2) and B(5, 2).



AB = _____

SCORE 4.0 Target Content

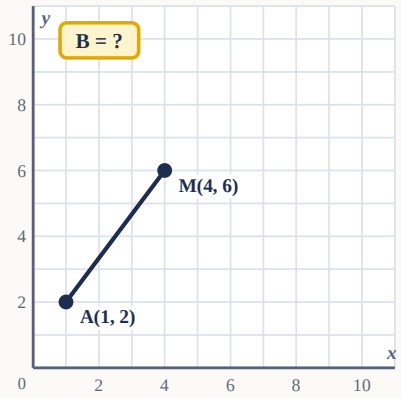
NC.M1.G-GPE.4 · NC.M1.G-GPE.5 · NC.M1.G-GPE.6 Partition segments, write parallel and perpendicular lines, and find perimeter.

• *I can find an endpoint, partition a segment, write equations of parallel and perpendicular lines, and find a perimeter.*

Solve each problem. Show your work on paper.

7. *Endpoint*

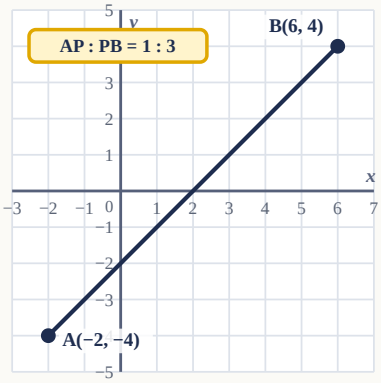
M(4, 6) is the midpoint of AB. A is (1, 2). Find B.



B = _____

8. *Partition*

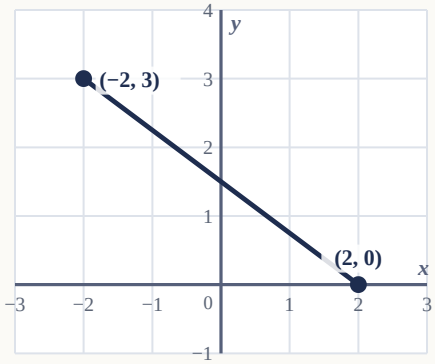
Point P is on AB so that AP : PB = 1 : 3. A is (−2, −4) and B is (6, 4). Find P.



P = _____

9. *Distance*

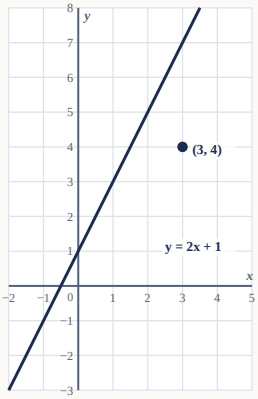
Find the distance between (−2, 3) and (2, 0).



d = _____

10. *Parallel line*

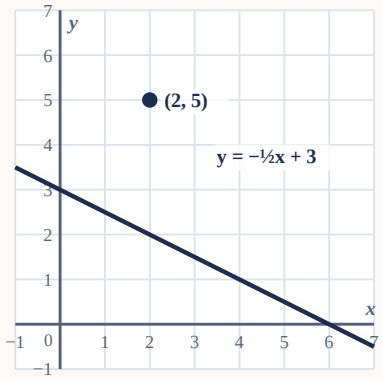
Write the equation of the line parallel to y = 2x + 1 that passes through (3, 4).



y = _____

11. *Perpendicular line*

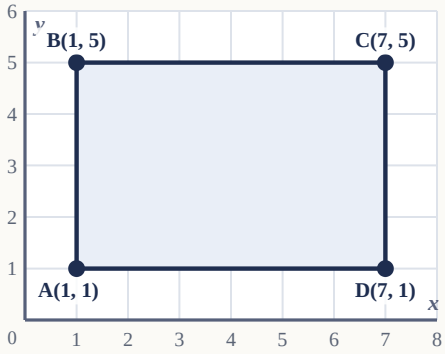
Write the equation of the line perpendicular to $y = -\frac{1}{2}x + 3$ that passes through (2, 5).



y = _____

12. *Perimeter*

Find the perimeter of rectangle ABCD.



P = _____

SCORE 5.0 Advanced Content

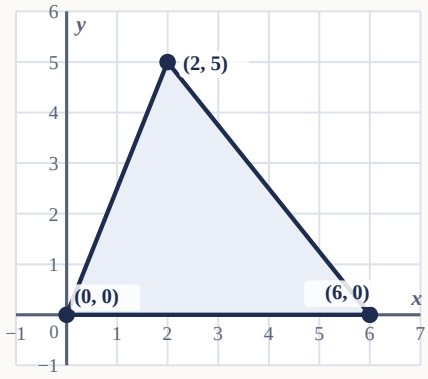
NC.M1.G-GPE.4 · NC.M1.G-GPE.5 · NC.M1.G-GPE.6 Use coordinates to prove facts about figures.

• *I can find area, classify triangles, prove a right angle with slopes, compare lines, and partition in any ratio.*

Solve each problem. Some answers are words.

13. *Area*

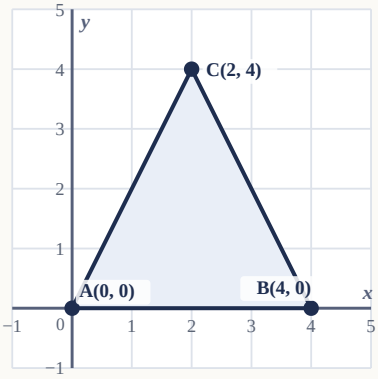
Find the area of the triangle with vertices (0, 0), (6, 0), and (2, 5).



Area = _____

14. *Classify*

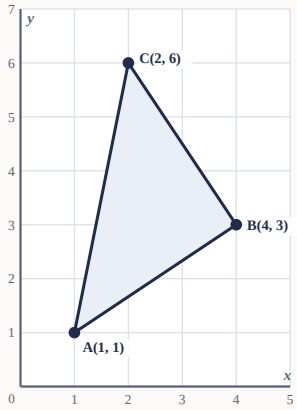
Classify triangle ABC by its sides: scalene, isosceles, or equilateral?



Answer: _____

15. *Right triangle?*

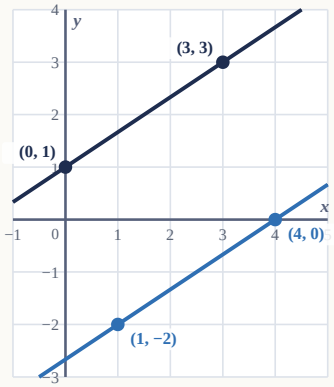
Is triangle ABC a right triangle? Write yes or no.



Answer: _____

16. *Compare lines*

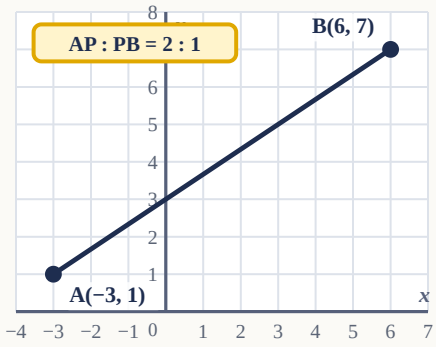
One line goes through (0, 1) and (3, 3). Another goes through (1, −2) and (4, 0). Are they parallel, perpendicular, or neither?



Answer: _____

17. *Partition*

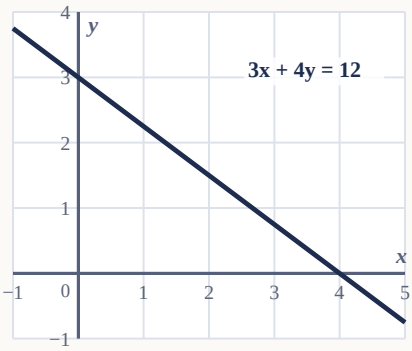
P is on AB so that AP : PB = 2 : 1. A is (−3, 1) and B is (6, 7). Find P.



P = _____

18. *Perpendicular slope*

What is the slope of a line perpendicular to 3x + 4y = 12?



m = _____